

2024

Time : 3 hours

Full Marks : 70

*Candidates are required to give their answers
in their own words as far as practicable.*

The figures in the margin indicate full marks.

*Answer **all** sections as directed.*

Section-A

Objective type Questions

Compulsory

1. Choose the correct option from the following multiple choice questions :

$$2 \times 10 = 20$$

(i) TCP is a _____ protocol.

- A) Connectionless
- B) Stateless
- C) Connection-oriented
- ☒ D) Secure

(ii) Which topology allows the network to continue working even if one link fails?

- A) Bus
- ☒ B) Star
- ☒ C) Mesh
- D) Ring

(iii) Which of the following is NOT a guided transmission medium?

- A) Twisted pair cable
- B) Coaxial cable
- C) Fiber optic cable
- ☒ D) Radio waves

(iv) Which transmission medium uses light signals for data transmission?

- A) Twisted pair
- B) Coaxial cable
- ☒ C) Optical fiber
- D) Satellite

(v) Which protocol provides reliable communication in the TCP/IP suite?

- A) UDP
- B) IP
- C) ICMP
- ☒ D) TCP

(vi) Which of the following is a connectionless protocol?

- A) TCP
- ☒ B) UDP
- C) FTP
- D) SMTP

(vii) How many sequence numbers are used in Stop and Wait ARQ ?

- A) 4
- B) 2
- C) 8
- D) Depends on packet size

(viii) In Go-Back-N, the sender can send _____ frames before requiring an ACK.

- A) Only one
- B) Any number
- C) Up to a window size
- D) Unlimited

(ix) If sequence number space is 8, what is the maximum window size in Selective Repeat?

- A) 4
- B) 8

C) 3

D) 7

(x) Which switching technique divides the message into smaller units?

- A) Circuit Switching
- B) Message Switching
- C) Packet Switching
- D) Token Switching

Section-B

Short Answer type Questions

Answer any **four** questions of the following :

5×4=20

2. Explain data link layer of OSI reference model.
3. How does stop and wait protocol ARQ help prevention duplication of frames At the receiver's side?

4. Differentiate between CSMA/CA and CSMA/CD.
5. Explain the causes of transmission impairments.
6. Why mesh topology is considered as a more reliable topology?
7. Write short note on packet switching.

Section-C

Long Answer type Questions

Answer any **two** questions of the following :

15×2=30

8. What is network protocol? Why do we need it? Explain the types of network protocols.
9. What is unguided transmission media? Describe different types of unguided media.

10. Explain the working of Go-Back-N ARQ with a suitable diagram.
11. The message 1101011001 is to be transmitted using the CRC with the generating polynomial $x^3 + x + 1$ to protect it from errors. What is the transmitted message?

_____ x _____